

2021 | (5) $M = 106 \text{ MeV}/c^2$, $\Delta t = 2.2 \mu\text{s}$



$$\Delta x = v \Delta t$$

$$v = \frac{100 \text{ km}}{2.2 \mu\text{s}} = \frac{100 \times 10^3}{2.2 \times 10^{-6}} \text{ m/s}$$

$$v > c$$

$$\frac{100 \times 10^3}{2.2} > 3 \times 10^8$$

* Lorentz transformation:

$$\begin{cases} \Delta x' = \gamma (\Delta x - v \Delta t) \\ \Delta t' = \gamma (\Delta t - v \frac{\Delta x}{c^2}) \end{cases}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

find v

$$\frac{\Delta x'}{\Delta t'} = v' = \left(\frac{\Delta x - v \Delta t}{\Delta t - \frac{v}{c^2} \Delta x} \right)$$

* proper time of the particle

$$c^2 \Delta \tau^2 = c^2 \Delta t^2 - \Delta x^2$$

proper time and coordinate time

$$\left(\frac{\Delta \tau}{\Delta t} \right)^2 = 1 - \left(\frac{\Delta x}{\Delta t} \right)^2$$

$$\left(\frac{\Delta \tau}{\Delta t} \right)^2 = 1 - \left(\frac{v}{c} \right)^2$$

$$1 - \left(\frac{v}{c} \right)^2 = \frac{1}{\gamma^2}$$

$$\left(\frac{\Delta \tau}{\Delta t} \right)^2 = \frac{1}{\gamma^2}$$

$$\gamma^2 \Delta \tau^2 = \Delta t^2$$

or

$$\Delta t = \gamma \Delta \tau$$

$$\Delta t = 2.2 \mu\text{s}$$

$$\Delta x = 100 \text{ km}$$

$$-\Delta \tau^2 = (\Delta t)^2 - \frac{(\Delta x)^2}{c^2}$$

$$= (2.2 \mu\text{s})^2 + \left(\frac{100 \text{ km}}{c} \right)^2$$

$$(2.2 \times 10^{-6})^2$$

$$\approx 4.84 \times 10^{-12}$$

$$1000 \times 10^3$$

$$+ \left(\frac{100 \text{ km}}{c} \right)^2$$

$$\left(\frac{1}{2} 10^{-3}\right)^2$$

$$10^{-6}$$

$$\frac{105}{108}$$

$$\frac{1}{10^3}$$

$$\gamma = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

* Try using lifetimes:

k = proper time
 $\hookrightarrow k = \frac{1}{\tau}$

$$N(t) = N_0 e^{-t/\tau}$$

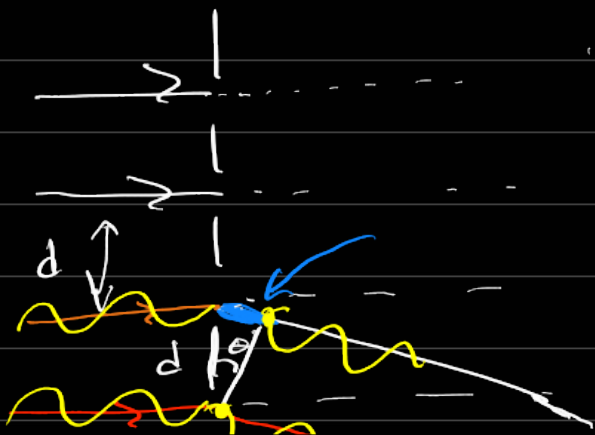
$$N(t) = N_0 e^{-t/\tau}$$

$$N(0) = N_0, \quad N(\Delta t) = N_0 e^{-\Delta t/\tau} = \frac{1}{2} N_0$$

$$\frac{\Delta t}{\tau} = \ln(2) = \gamma \rightarrow \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

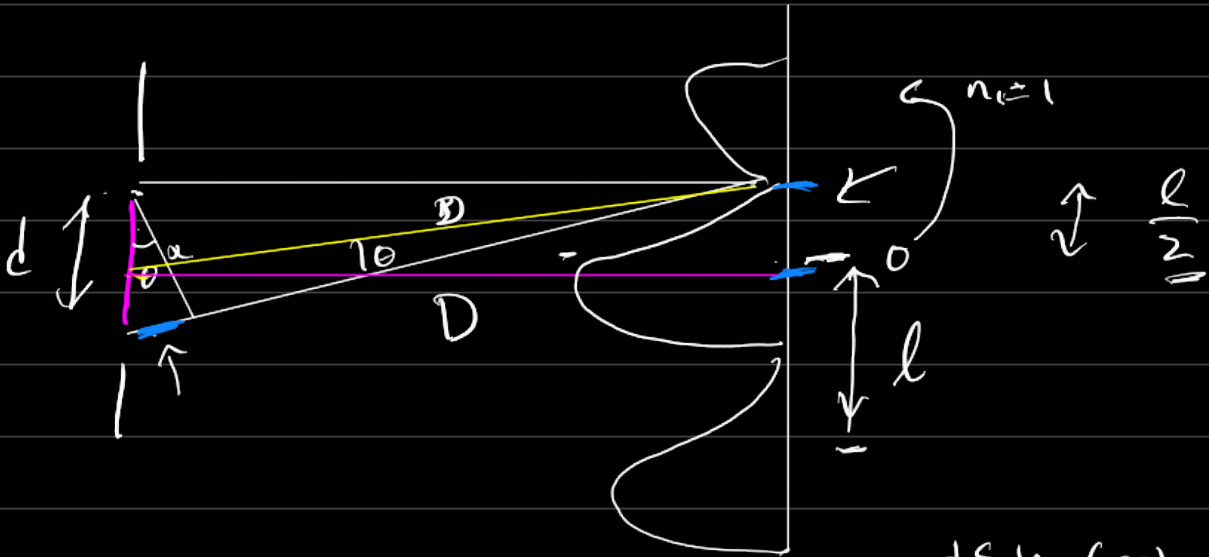
Solve for v

$$E = \gamma m c^2$$



$$ds \sin \theta = n \lambda$$

1



$$\frac{l}{2} = \frac{\lambda D}{2d}$$

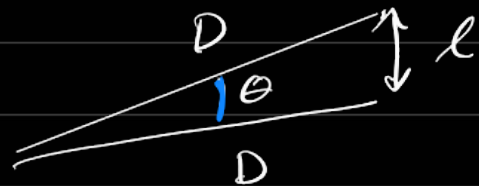
$$l = \frac{\lambda D}{d}$$

$$d = \frac{\lambda D}{l}$$

$$d \approx \frac{\lambda}{\theta}$$

$$d \sin(\theta) = \lambda$$

$$\sin(\theta) = \frac{l/2}{D}$$



$$D\theta \approx l$$

$$\theta \approx \frac{l}{D}$$

$$\theta = 2 \text{ arc sec} \rightarrow \text{radians}$$

